# TAN YI CHERNG

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**Electrical & Electronics Engineering** undergraduate at University of Southampton driven by a passion for technology, Robotics, and innovative problem-solving. Actively seeking opportunities across a range of STEM field to expand my knowledge in embedded systems, AI, and IoT, while collaborating with others to solve real-world challenges.

## EDUCATION

**University of Southampton (UoS)**

OCT 2022 – Present
**MEng Electrical & Electronics Engineering (3rd year)**
- Academic Excellence Progression Scholarship (25% scholarship for 1st year)
- 1st Year Part Average: **81.3%**
- 2nd Year Part Average: **71.5%**
- Anticipated graduation in the year **2026**.

JUL 2021 – JUN 2022
**Foundation in Engineering**
- High Achiever Scholarships (50% scholarship for Engineering Foundation Year)
- Part Average: **85.7%**

Modules Highlight:

*Electronic Circuits, Programming, Digital System and Microprocessors, Advanced Programming, Electronic Systems, Solid State Devices, Electrical Materials and Fields, Computer Engineering, Power Electronics and Drives, Electrical and Electronics Engineering Design, Control and Communications, Digital System and Signal Processing, Power Circuit and Transmission, Applied Electromagnetism, Mathematics for Electronic and Electrical engineering, Digital IC and System Design, Embedded Network System, Digital Control System Design, Security of Cyber-Physical System, Engineering Management and Law.*

## EXPERIENCE

JUN 2024 – SEP 2024
**Research Internship,** UoS Malaysia Campus
- A research opportunity by Dr Chin Vun Jack in **designing an Internet-Enabled I-V Curve Tracer** utilizing Cuk converter, focused on creating an affordable solution for I-V curve tracing with remote monitoring.
- Utilized **Raspberry Pi 4** for data acquisition, integrated with a **TFT display** and **webserver connectivity**, providing real-time data access via Wi-Fi and cloud-based solution.
- Successfully developed a prototype for **RM140 (£25)**, achieving an RMSE of 0.28A and RRMSE of 6.5%.
- Submitted a **conference paper** to **Gen-CITy 2024**, University of Southampton Malaysia.

SEP 2023 – DEC 2023
**Part-time Internship**, Carbon GPT
- A start-up company specializing in AI-driven, cloud-based **carbon footprint management SaaS platform** designed to aid organizations in managing their carbon emissions and sustainability efforts effectively.
- Engaged as a **software development intern**, collaborating with cross-functional teams in **developing and implementing key features** for the Carbon GPT SaaS platform.
- Actively contributed to **UI/UX design** enhancement and **systematic troubleshooting** efforts.

JUN 2023 – AUG 2023
**Research Internship**, UoS Malaysia Campus
- A research project led by Dr. Suan Hui Pu in **designing an electronic stethoscope** with low-cost MEMS microphones to achieve low-cost, ubiquitous sensing for improving respiratory healthcare.
- Worked on the **development of a web-based application** tailored for research on the electronic stethoscope.
- Contributed to the publication of a **conference paper** at the **2023 IEEE International Conference on SENNANO.**

APR 2023 – APR 2024
**President**, Robotics Club of UoS Malaysia Campus
- Lead the **Robotics Club** consisting of 80 members and 8 committees in **planning and organising robotics-related workshops and events**.
- **Organised eight** robotics-related events for the students in the 2023/2024 terms.
- Members count increased by **23%** & Revenue increased by **62%.**
- Won the **2024 Best Society Award.**

## PROJECT HIGHLIGHTS

AUG 2024 – SEP 2024

### Automated Popiah Skin Making Machine

- A project to develop a prototype machine capable of **automating the process of making Popiah Skin**.
- **Led the electrical and electronics integration**, collaborating closely with the mechanical team.
- Design and implement a **power supply** to convert 240V AC to power eight heating elements, while providing a regulated 12V DC for the control circuitry.
- Built a **control system** using an **ESP32** microcontroller with **FreeRTOS** to regulate temperature control and motor speed.

FEB 2024 – MAY 2024

### Internet-Enabled Smart Meter Design

- **Second year design project** to design a smart meter from scratch in a team of five.
- Designed a **waterproof smart meter housing** using **SolidWorks CAD** software and manufactured it through 3D printing.
- Designed two **power supply unit** (SMPS & transformer-based) to power the system's circuitry.
- Designed an **interface circuitry** for seamless interaction with a virtual microgrid emulator.
- Designed an **algorithm to maximize the survival rate** of the population within a testbed.
- Integrated **internet connectivity** for **real-time data collection**, **cloud-based storage** and displaying it on a **website** hosted live on Render.
- Implemented an **AI chatbot** using **OpenAI GPT-3.5 turbo 0125** capable of accessing data from the database through the **function calling** feature.

OCT 2023 – MAY 2024

### Integrated Circuit Design Project

- Led a team of four in **design**, **layout**, and **testing** of **integrated circuit (IC)** composed of three digital modules: Ring Oscillator; 9-bit Sequence Detector, and Asynchronous Serial Interface.
- Utilized tanner tools such as **S-Edit**, **L-Edit** and simulation tools like **CircuitVerse** and **ModelSim**.
- The IC was developed to achieve the design rules specified by Taiwan Semiconductor Manufacturing Company (RSMC).
- Experienced the **full IC design process**, including **functional verification**, and **fabrication constrains**.

JUL 2023 – AUG 2023

### Quadruped Spider Robot

- Designed a four-legged spider robot from scratch as a part of the Robotics Club workshop.
- Powered by Cytron Maker Uno with 8 servos that are **wirelessly controlled** with a mobile application via **Bluetooth**.
- Designed custom parts and support frames for the robot using **SolidWorks CAD** software.
- Parts are manufactured and mass-produced using **PLA 3D printing filament** and **laser cutting** of plywood on the campus.

MAY 2023 – JUN 2023

### Paediatric Malnutrition Early Detector (PaeMed)

- Won the **First Runner-up** of INVENCMAX 3.0 by IEM-UTM SS
- Won the **Second Runner-up** of Smart City Hackathon 2023
- Designed a pair of **weight sensing insoles** and a **height measuring cap** to combat the ever-rising rate of malnutrition among children through early detection.
- The products are wirelessly **connected to an AI powered mobile application** via **Bluetooth** for child growth analysis and monitoring to detect malnutrition among children.
- Well-aligned with **SDG goals 2 and 3**, target 2.2 (malnutrition) and target 3.4 (non-communicable diseases).

MAR 2023 – MAY 2023

### RFID PCB Layout and Design

- Utilizing **Autodesk EAGLE CAD** software for the **PCB design** and exporting for **PCB fabrication**.
- Involved creating **custom footprints**, **capturing schematics**, **laying out a two-layer PCB for a mixed-signal circuit design**, and **testing the fabricated PCB**

SEP 2022 – OCT 2022

### Flappy Bird Replica Game

- Using Ilmatto microcontroller board (ATmega644p onboard), TFT LCD display, and a button to make a **simple game console** that runs a replica version of the flappy bird game.
- The game was coded using pure **C language**.

## SKILLS

*C, C++, Python, HTML, JavaScript, CSS, Arduino, Raspberry Pi, FreeRTOS, FPGA, SystemVerilog, Flask Framework, Design for Test (DFT), VLSI, Digital IC Design, Circuit Design, Layout Design, Large Language Model (LLM), Printed Circuit Board (PCB), Robotics, Embedded System, Power Electronics, Computer Architecture, Google Firebase, SolidWorks CAD, MATLAB/ Simulink, ModelSim, Intel Quartus Prime, NI Multisim, SPICE software, EAGLE CAD, S-edit, L-edit.*